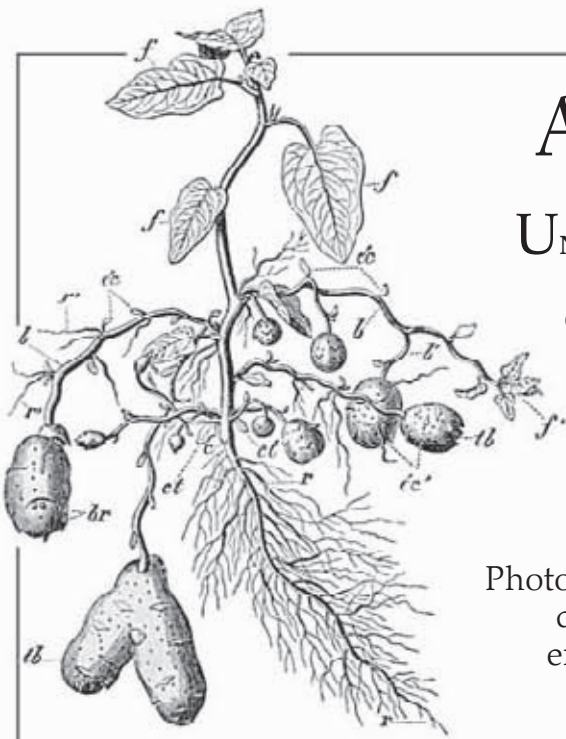


# A light diet

UNLIKE ANIMALS, PLANTS DO NOT NEED to find food, because they can make it for themselves. The key to this is a substance called chlorophyll, which makes leaves green. Plants use chlorophyll to trap the energy in sunlight. Once they have done this, the plants can put the energy to work. They use it to make water combine with carbon dioxide, a gas that they get from the air. The result is glucose—an energy-rich food sugar that plants use to grow. This process is called photosynthesis, which means putting together with light. Photosynthesis works like a bonfire in reverse. While a bonfire breaks down living things and gives out energy, photosynthesis takes in energy and uses it to build up the living parts of plants. Thanks to photosynthesis, plants make billions of tons of food every year. Without it, most of the living world would soon come to a halt, because plants make the food that other living things need to survive.



## UNDERGROUND STORAGE

Potatoes are swollen underground stems, known as tubers, which store the food produced by photosynthesis. This food, in the form of starches, provides the young shoots that develop from buds on the tuber with enough fuel to enable them to grow quickly. Potatoes also provide an important source of human food, and they have been bred to produce bigger tubers.

*Leaves produced in the dark have little chlorophyll, leaving them pale, or etiolated*

*Potato tuber kept in the dark for six months*

*Stems grow upward against gravity*

*Each stem is produced by a small bud called an eye*

## A PLANT WITHOUT LIGHT

This potato has spent six months with very little light—conditions that would kill many plants. Because it has been in almost complete darkness, the young potato has not been able to produce any food by photosynthesis. However, it has survived and has even produced some roots and shoots, which have drawn on the food reserves stored by the parent plant during the previous year's growth. The parent potato plant used the Sun's energy to make food, which it stored in its tubers, mostly in the form of grains of starch. The young potato plants use the starch to release energy for growth.

*Roots that grow from stems are called adventitious roots*

*Tuber begins to shrink as its food stores are used up*

